Project Specifications

Description:

While in-class exercises and homework assignments can develop your understanding and skills, they cannot fully prepare you for working with data "in the wild." The goal of the final project is to apply your skills in data wrangling with R to real-world data in order to explore a problem facing business or society. The project gives each team the freedom to expand on the concepts learned in lecture and lab and work in a domain that they are passionate about.

- This project may be completed individually or in groups of one, two, or three. You do not need to be in the same lab section to be partners. If you work in groups of 2 or 3, only ONE person should turn in the deliverables. Make sure you include all member's names on the work.
- The project must leverage at least two sources of real (not simulated) data. The datasets must be able to be integrated into a single data frame (vertically or horizontally merged). At least one dataset must be obtained by you via web scraping and/or APIs. The other can be a pre-existing dataset downloaded from a site like Kaggle or Data.gov. (Some possible data sources are listed in "Open Data Sources")
- Groups will perform descriptive analytics using the integrated data, including calculating summary statistics, hypothesis testing, fitting models, and creating visualizations. While you are strongly encouraged to generate your own topic, some example topics include:
    - Analyze Congress members' voting records considering the population demographics of their home states (NY Times API, Census data)
    - Analyze the social media fan base of college sports teams with consideration of the makeup of their student body (Twitter API, Education data)
    - Rank a group of movies by box office revenues, accounting for inflation (Wikipedia data, Inflation data)

Grading:

The final project is worth 100 points total (17% of your final grade). The points are assigned to each deliverable as follows: Submissions will be assessed based on the following criteria:

- Novelty: What interesting data sources are used? What type of unique analysis and/or insights does your project provide?

- Difficulty: How challenging is the implementation of your project? How difficult is it to retrieve, process, and integrate the datasets? Does your code extend beyond skills covered in course materials?
- Accuracy: Has the data been cleaned and integrated in a way that does not introduce errors or bias? Are the visualizations and analytical methods appropriate for the data and/or research questions?
- Clarity: Does your report clearly describe the motivation, research questions, data sources, analyses, and results of your project? Is the report written in a professional (error-free) style?

Item	Points
Proposal (March 22nd)	25
Check-In (April 12th)	25
Project Report, Data & Code (May 3rd)	50

Project Proposal:

Each team must submit a proposal to ICON by Friday, March 22nd @ 11:59pm. The proposal should include:

- Introduction: Provide background information on the context so that a non-expert can understand. Describe the problem that your descriptive analysis is meant to explore. Make sure to include links/citations for any facts and figures that are not common knowledge. For example: “The University of Iowa is a college in Iowa City” is common knowledge. “Jeff Bezos has a current net worth of $140 billion” is a fact that should be cited.
- Data: Clearly describe the source of your data. By the time of the project proposal, you should have identified at least one of your data sources. Include a description of how you plan to collect at least one additional data source (e.g., scrape data from a website, collect data from a cryptocurrency API). Make sure to include a link/citation for all data sources. Create a “data dictionary,” a table that lists fieldnames, data types, and descriptions (example below).

Example data dictionary (college football data)

Field	Type	Description
College	Text	College name
City	Text	College city
State	Text	College state
Division	Text	Football division (e.g., FBS, FCS)
Conference	Text	Football conference (e.g., PAC12)
AP	Numeric	Team preseason AP poll ranking
WP	Numeric	Team winning percentage over last 5 years
Bowl	Logical	Team played in postseason bowl in previous season?
Draft	Numeric	Number of players taken in NFL draft over last 5 years

- Proposed Analysis: Present at least three potential research question(s). For some questions, you may have a hypothesis (expected result) in mind, but others may simply be exploratory. The final project must include multiple methods from the course (e.g., a combination of summary statistics, visualization, and statistical tests/models). One question may be answered from a single data set. The other questions must be answered from your combined data sets.

Project Check-In:

Each group will receive feedback on their project proposal. After that point, each group must submit an updated check-in with progress on the report showing scraped, downloaded, and cleaned data by Friday, April 12th, 2024.
More details on the check-in will be released after Spring Break.

Project Report:

Your final report should be a clearly and professionally written document that includes all of your design and implementation details as well as data analysis and results. Specifically, you need to organize your report into four sections as follows (can have additional sub-sections):

- **Introduction:** Provide background information on the domain and motivation for study (why it's important). Also, introduce your planned analysis and initial hypotheses/intuition. (1 page)
- **Data:** Describe the sources of all your data (including citation/link). Additionally, describe how you obtained, wrangled, cleaned, and integrated the data. Create a data dictionary for the final, merged dataset (2-3 pages)
- **Analysis:** Present your research questions and results. You should interpret the results of your analysis considering your initial hypotheses and the broader context. Be careful not to extrapolate or over-generalize (4-6 pages including graphs) You should have 3-4 research question groups. Use descriptive statistics, modeling, hypothesis testing, and importantly visualizations to answer your questions.
- **Conclusion:** Briefly present your overall conclusions, limitations, and suggestions for future work. (1 page)
- **The report will be between 8-11 pages 1.5 spaced, 12 point font. Include graphs within your analysis. You may have a maximum of 8 additional graphs in an appendix that you reference in your report.**
- Each group (1 group member submits with all names on report) must submit the project report (.docx or .pdf), R scripts (.R), and R data files (.rdata). The R data files are the saved dataframes (both individual and merged. Submit to ICON by 11:59 pm on Friday, May 3rd, 2024.

Project Data/Code:

All work for the project must be completed in R/RStudio, and all code should be provided in the form of R scripts. In order to ensure that your work is portable and reproducible, it is highly recommended that you separate your work into multiple scripts.

For example, you could create one script for Web scraping, and then save the data in a file. Then a second script for data integration, saving the final, merged data, and a third script containing all descriptive analysis. Ensure your scripts are appropriately named so we can determine the run order.

Each group must submit one or more R data file(s) (.rdata) with all data required to replicate your results. That means that your group must submit the either the source datasets and/or the final, merged dataset.